

Week 15 Capstone — Client Brief (Template + Worked Example)

Your team writes its own client brief for the dataset you chose — there's no instructor-provided brief this time. Write it **before** starting the lab notebook; it sets the context for your executive summary memo (Section E) and your Capstone Report.

A good brief has three parts: a named client/stakeholder, a short paragraph establishing why they care about your data, and 2–3 specific questions they want answered. Your questions should map onto what your analysis can actually answer — the relationship between your predictor and outcome (regression), a difference between your two groups (hypothesis test), and the range of uncertainty in your outcome (Monte Carlo).

Template

Client: [A named, plausible stakeholder — a real job title at a real or realistic organization]

[2–4 sentences establishing the client's situation: what they're responsible for, why this data matters to them, and what decision or budget question is riding on the answer. End with 2–3 numbered questions the client wants answered — write these to be answerable by your regression, hypothesis test, and Monte Carlo results.]

Your dataset: [filename or source] — [row count] observations, [what each row represents], [time period/location if relevant].

Worked Example — Hospital (using a legacy building-energy dataset)

Client: Facility Manager, Banner University Medical Center (hypothetical 800-bed acute-care hospital, Tucson)

Your hospital runs 24 hours a day, 365 days a year. Cooling is not optional — patient rooms, operating theaters, and pharmacy storage all have strict temperature requirements. Last fiscal year, cooling electricity accounted for 38% of the facility's total energy bill. The CFO has asked for a data-driven analysis to answer three questions: (1) How strongly does outdoor temperature drive cooling load? (2) Is there a measurable difference in cooling during standard business hours versus overnight? If so, what does the coefficient mean operationally for a 24/7 facility? (3) What is the expected annual cooling cost, and what is the realistic range if electricity rates fluctuate?

Your dataset: `Week15_Hospital_Dataset.csv` — 8,760 hourly observations, Tucson climate, 2023. Predictor (X) = outdoor temperature; outcome (Y) = cooling energy; group = business hours vs. non-business hours.

Tips

- **Pick a client who would realistically pay for this analysis.** “A city engineer deciding whether to add a signal at an intersection” is more specific and more gradeable than “someone interested in traffic.”
- **Write questions your methods can actually answer.** Don’t ask “should we build a new facility?” (too broad) — ask “how much does X predict Y, and how confident are we?” (matches your regression + Monte Carlo directly).
- **If your grouping variable produces a counterintuitive result** (e.g., a hotel dataset where evening peak occupancy makes daytime “business hours” look like a *lower*-cooling period), don’t avoid it. Write a question that specifically asks your team to explain it. That’s usually the most interesting part of the analysis.